

Software Requirements Specification (SRS) for Smart Yoga Mat

Part A: Introduction

1.1 Purpose of the Document

The purpose of this document is to define the requirements for developing a **Smart Yoga Mat** integrated with **AI technology**. The system will assist users in performing yoga correctly by tracking posture, providing real-time feedback, and monitoring overall wellness.

1.2 Scope of the Document

The **Smart Yoga Mat** will:

- Detect yoga poses using embedded sensors and AI algorithms.
- Provide real-time feedback on posture correction to ensure correct alignment and prevent injury.
- Track performance and progress, including metrics such as **calories burned, time spent, and pose accuracy**.
- Integrate with **mobile applications** for visualization, progress tracking, and personalized training plans.
- Support both **beginner and advanced users** by providing dynamic feedback and personalized recommendations.

1.3 Overview

This system combines **IoT (Internet of Things)** technology (sensors embedded in the mat) with **AI (Artificial Intelligence)** for pose detection and real-time feedback. The mat ensures correct posture, prevents injury, and enhances the user's yoga practice efficiency by providing personalized, actionable insights.

Part B: General Description

2.1 System Overview

The system consists of a **smart yoga mat** with built-in **pressure sensors, accelerometers, and gyroscopes** that collect data from the user's movements. The data is processed using AI algorithms to detect and classify various yoga poses in real-time. Based on pose recognition, the system provides feedback and tracks progress.

- **Sensors on the mat** will gather real-time data on the user's movements and posture.
- **AI algorithms** will process this data to accurately detect yoga poses.

- Feedback will be provided through the **mobile application** and can include **visual** or **audio cues**.
- The system will store user progress and suggest **personalized training plans** to improve practice.

2.2 User Characteristics

- **Yoga Practitioners:** Both **beginner and advanced** users who want to improve their practice, track performance, and receive feedback on posture.
- **Fitness Coaches/Instructors:** Yoga instructors using the system for student feedback and to personalize sessions.

2.3 Assumptions and Dependencies

- The system assumes users have access to a **smartphone** or **tablet** for the mobile application.
 - The system will rely on **Bluetooth** or **Wi-Fi** connectivity to sync data with the app.
 - The smart mat will operate effectively in **indoor environments** with stable **Wi-Fi** or **Bluetooth** connectivity.
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Part C: Functional Requirements

3.1 Pose Detection and Classification

- The system must detect and classify **yoga poses** in real-time with an accuracy of at least **90%**.
- The system will support common yoga poses such as **Downward Dog**, **Tree Pose**, **Warrior**, and others.

3.2 Real-Time Feedback

- Provide **corrective feedback** for posture alignment through visual cues (on the app), **auditory cues** (voice assistant), and **haptic feedback** (vibration on the mat).

3.3 Performance Tracking

- Track and record session **statistics** including **duration**, **calories burned**, and **pose accuracy**.
- Display metrics on a **dashboard** in the mobile app for visualization.

3.4 User Profile and Personalization

- Users can create and **personalize** profiles, allowing the system to tailor feedback and recommendations.

- Support for multiple user profiles per app to enable family or group use.

3.5 Data Syncing

- Sync data with a **mobile application** in real-time via Bluetooth or Wi-Fi.
 - Allow for **offline mode** where users can still use the mat, with data syncing once the internet connection is available.
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Part D: Interface Requirements

4.1 Hardware Interface

- The **smart yoga mat** will contain **pressure sensors**, **accelerometers**, and **gyroscopes**.
- The system will communicate with the **mobile app** via **Bluetooth** or **Wi-Fi**.
- Charging port: USB-C or **wireless charging**.

4.2 Software Interface

- **Mobile Application:** The app will be compatible with both **Android** and **iOS** platforms.
 - It will include a **dashboard** to show progress, statistics, pose library, and recommendations.
 - The app will have a **user-friendly interface** to manage profiles and settings.

4.3 User Interface

- The mobile app will feature an **interactive dashboard** that will display:
 - **Posture analysis:** Showing pose accuracy and posture corrections.
 - **Progress tracking:** Displaying user history and performance data.
 - **Pose library:** A catalog of yoga poses, including instructions and visual guides.

4.4 Voice/Audio Interface

- The system will offer **real-time corrective guidance** through a **voice assistant**.
 - Voice prompts will guide users on correcting posture and give feedback during practice.
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Part E: Performance Requirements

5.1 Response Time

- The system should provide real-time feedback with **latency of less than 1 second** after pose recognition.

5.2 Pose Recognition Accuracy

- The system must achieve **pose recognition accuracy of $\geq 90\%$** for common yoga poses.

5.3 Battery Life

- The smart yoga mat should have a battery life of at least **8 hours** of continuous use on a full charge.

5.4 Data Storage

- The system should support **up to 1000 stored sessions** per user, allowing for long-term tracking of progress.
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Part F: Design Constraints

6.1 Hardware Constraints

- The mat should be **lightweight** and **foldable**, allowing for easy portability and storage.
- The **pressure sensors** should be durable, flexible, and waterproof to withstand regular use and cleaning.

6.2 Software Constraints

- The mobile application should work seamlessly on **Android (version 8.0 and above)** and **iOS (version 12.0 and above)**.
- The system should perform well on mobile devices with **Bluetooth 4.0 or higher**.

6.3 Environmental Constraints

- The system must work effectively in **indoor environments** where Wi-Fi or Bluetooth signals are stable.
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Part G: Non-Functional Attributes

7.1 Usability

- The system should be **easy to use** for all age groups and skill levels.
- The mobile app should have a **simple interface**, ensuring that even beginners can navigate it easily.

7.2 Reliability

- The system should function consistently for long durations ($\geq 99\%$ uptime), ensuring reliability during yoga sessions.

7.3 Security

- **User data** should be **encrypted** during transmission and storage to ensure privacy.
- The app should allow users to **delete** their account and data if desired.

7.4 Scalability

- The system should support multiple users on a single mobile app, allowing families or groups to use the same device.

7.5 Maintainability

- The system should allow for **software updates** via the mobile app, enabling new features or bug fixes without requiring user intervention.

Part H: Preliminary Schedule and Budget

8.1 Schedule

- **Research & Design:** 1 month
- **Hardware Development & AI Model Training:** 3 months
- **System Integration & Testing:** 2 months
- **Deployment & Launch:** 1 month

Total Estimated Development Time: 7 months

8.2 Budget (Estimated)

- **Hardware** (sensors, mat, connectivity): ₹15,000–20,000
- **Software Development:** ₹2,50,000
- **AI Model Training:** ₹50,000
- **Marketing & Distribution:** ₹1,00,000

Total Estimated Budget: ₹4,20,000